

Practice Questions for Shadow Prices

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December 31, 2024

1 Questions

1. In the context of linear programming, what is the economic interpretation of a shadow price, and how does it relate to the optimal solution?
 - a. The shadow price represents the marginal cost of producing one more unit of the objective function.
 - b. The shadow price represents the marginal profit obtained by increasing the availability of a resource by one unit.
 - c. The shadow price represents the amount by which the optimal objective function value would change if a constraint's right-hand side were increased by one unit.
 - d. The shadow price represents the rate of change of the objective function with respect to a change in a decision variable.
 - e. The shadow price represents the minimum amount by which a constraint's right-hand side must be increased to change the optimal solution.
2. Suppose a linear programming problem has an optimal solution. If a non-basic variable has a shadow price of 0, what does this imply about the corresponding constraint in the dual problem?
 - a. The constraint is binding (holds with equality) in the optimal dual solution.
 - b. The constraint is non-binding (holds with strict inequality) in the optimal dual solution.
 - c. The constraint is redundant and can be removed without affecting the optimal solution.
 - d. The constraint is infeasible in the dual problem.
 - e. The shadow price of 0 provides no information about the corresponding dual constraint.
3. How are shadow prices used in sensitivity analysis to assess the impact of changes in resource availability on the optimal objective function value?
 - a. Shadow prices directly indicate the new optimal objective function value after a resource change.
 - b. Shadow prices provide an approximation of the change in the optimal objective function value for small changes in resource availability.
 - c. Shadow prices are irrelevant to sensitivity analysis; only changes in the decision variables matter.
 - d. Shadow prices only apply to changes in the objective function coefficients, not resource availability.
 - e. Shadow prices indicate the exact change in the optimal objective function value only for large changes in resource availability.

4. What is the relationship between shadow prices and the dual variables in a linear programming problem?
 - a. Shadow prices are the negative of the dual variables.
 - b. Shadow prices are the absolute values of the dual variables.
 - c. Shadow prices are equal to the dual variables.
 - d. There is no direct relationship between shadow prices and dual variables.
 - e. Shadow prices are the reciprocals of the dual variables.
5.
 - a. Explain the concept of shadow prices (dual variables) in linear programming and their economic interpretation.
 - b. How are shadow prices used in sensitivity analysis to assess the impact of changes in resource availability on the optimal objective function value?
 - c. Suppose a linear programming problem has an optimal solution. If a non-basic variable has a shadow price of 0, what does this imply about the corresponding constraint in the dual problem?
 - d. What is the relationship between shadow prices and the dual variables in a linear programming problem? Provide a concise mathematical explanation.
 - e. Describe a scenario in a real-world application where understanding shadow prices would be crucial for effective decision-making. Explain how the shadow prices would inform the decision.
 - A. Shadow prices represent the marginal value of increasing the right-hand side of a constraint by one unit, holding other constraints constant. They are equal to the dual variables at the optimal solution.
 - B. Shadow prices are used to determine the optimal solution to the primal problem. They are calculated by solving the dual problem directly.
 - C. Shadow prices are only relevant for basic variables in the optimal solution. They indicate the rate of change of the objective function with respect to changes in the basic variables.
 - D. Shadow prices are independent of the dual variables. They represent the cost of violating a constraint.
 - E. Shadow prices are calculated by performing a sensitivity analysis on the primal problem. They indicate the range of values for which the optimal solution remains unchanged.
6. In the context of shadow prices, what is the economic interpretation of a shadow price of 0 for a constraint in a linear programming problem?
 - a. The constraint is binding, and a small change in its RHS will significantly impact the optimal objective function value.
 - b. The constraint is non-binding, and a small change in its RHS will not affect the optimal objective function value.
 - c. The constraint is binding, and a small change in its RHS will not affect the optimal objective function value.
 - d. The constraint is non-binding, and a small change in its RHS will significantly impact the optimal objective function value.
 - e. The shadow price of 0 is meaningless and provides no information about the constraint.

7. Consider a maximization linear programming problem. A non-basic variable has a shadow price of 5. What does this indicate?
 - a. Increasing the resource associated with this variable will increase the objective function value by approximately 5 units per unit increase.
 - b. Decreasing the resource associated with this variable will increase the objective function value by approximately 5 units per unit decrease.
 - c. The variable is already at its optimal level, and changes will not affect the objective function.
 - d. The variable's current value is 5.
 - e. The shadow price is irrelevant to the non-basic variable.
8. In a linear programming problem, the shadow price for a particular constraint is 2. What does this mean?
 - a. The optimal objective function value will decrease by 2 units if the right-hand side of the constraint is increased by 1 unit.
 - b. The optimal objective function value will increase by 2 units if the right-hand side of the constraint is increased by 1 unit.
 - c. The optimal objective function value will remain unchanged if the right-hand side of the constraint is increased by 1 unit.
 - d. The optimal solution will remain unchanged if the right-hand side of the constraint is increased by 1 unit.
 - e. The shadow price is irrelevant to the optimal objective function value.
9. Suppose a linear programming problem has an optimal solution. If a constraint has a shadow price of 0, what does this imply?
 - a. The constraint is binding, and the optimal solution lies on the constraint boundary.
 - b. The constraint is non-binding, and the optimal solution lies within the feasible region, away from the constraint boundary.
 - c. The optimal solution is infeasible.
 - d. The problem is unbounded.
 - e. The shadow price is undefined for this constraint.
10. In a linear programming problem, a shadow price of 0 for a particular constraint implies what about the associated resource?
 - a. The resource is being fully utilized in the optimal solution.
 - b. The resource is not being fully utilized in the optimal solution, and increasing its availability would not improve the objective function value.
 - c. The resource is scarce and increasing its availability would significantly improve the objective function value.
 - d. The optimal solution is degenerate.
 - e. The problem is infeasible.

2 Answers

1. In the context of linear programming, what is the economic interpretation of a shadow price, and how does it relate to the optimal solution?
 - a. The marginal cost of producing one more unit of the objective function is not directly represented by the shadow price. Shadow prices relate to constraints, not the objective function directly.
 - b. While shadow prices relate to resource availability, they represent the *increase* in the objective function value, not necessarily profit. The interpretation depends on whether the problem is a maximization or minimization problem.
 - c. A shadow price represents the marginal value of relaxing a constraint by one unit. It indicates how much the optimal objective function value would improve if the constraint's right-hand side were increased by one unit. This reflects the value of additional resources.
 - d. The rate of change of the objective function with respect to a decision variable is related to reduced costs, not shadow prices. Shadow prices are associated with constraints.
 - e. The minimum amount by which a constraint's right-hand side must be increased to change the optimal solution is related to the range of feasibility, not the shadow price itself.
2. Suppose a linear programming problem has an optimal solution. If a non-basic variable has a shadow price of 0, what does this imply about the corresponding constraint in the dual problem?
 - a. If the shadow price is 0, the corresponding dual constraint is not binding. A binding constraint would have a non-zero shadow price.
 - b. A shadow price of 0 for a non-basic variable in the primal problem implies that the corresponding constraint in the dual problem is non-binding. This is a direct consequence of complementary slackness.
 - c. A shadow price of 0 doesn't necessarily mean the constraint is redundant. It simply means it's not actively limiting the optimal solution.
 - d. An infeasible constraint would typically lead to an infeasible dual problem, not a shadow price of 0 for a specific constraint.
 - e. The shadow price of 0 is directly related to the corresponding dual constraint via complementary slackness.
3. How are shadow prices used in sensitivity analysis to assess the impact of changes in resource availability on the optimal objective function value?
 - a. Shadow prices give an *approximation*, not the exact new optimal value, especially for larger changes in resource availability.
 - b. Shadow prices provide a linear approximation of the change in the optimal objective function value for small changes in resource availability. For larger changes, the approximation may become less accurate, and the optimal solution's basis might change.
 - c. Shadow prices are fundamental to sensitivity analysis, specifically for assessing the impact of changes in resource availability (right-hand side values of constraints).
 - d. Shadow prices are directly related to resource availability (the right-hand side of constraints), not just objective function coefficients.
 - e. Shadow prices provide a better approximation for *small* changes. For large changes, the optimal basis might change, invalidating the linear approximation.

4. What is the relationship between shadow prices and the dual variables in a linear programming problem?
 - a. Shadow prices are not the negative of the dual variables. The sign depends on whether the primal problem is a maximization or minimization problem.
 - b. Shadow prices are not simply the absolute values of the dual variables. The sign is meaningful and reflects the direction of the change in the objective function.
 - c. Shadow prices are numerically equivalent to the optimal values of the dual variables. They both represent the marginal value of increasing a resource (relaxing a constraint).
 - d. There is a direct and crucial relationship: shadow prices are the optimal dual variables.
 - e. Shadow prices are not the reciprocals of the dual variables. There's a direct numerical equivalence.
5.
 - a. Explain the concept of shadow prices (dual variables) in linear programming and their economic interpretation.
 - b. How are shadow prices used in sensitivity analysis to assess the impact of changes in resource availability on the optimal objective function value?
 - c. Suppose a linear programming problem has an optimal solution. If a non-basic variable has a shadow price of 0, what does this imply about the corresponding constraint in the dual problem?
 - d. What is the relationship between shadow prices and the dual variables in a linear programming problem? Provide a concise mathematical explanation.
 - e. Describe a scenario in a real-world application where understanding shadow prices would be crucial for effective decision-making. Explain how the shadow prices would inform the decision.
 - A. Shadow prices, also known as dual variables, represent the marginal increase in the optimal objective function value resulting from a one-unit increase in the right-hand side of a constraint, assuming all other parameters remain constant. This aligns with the economic interpretation of the dual variables as the marginal value of a resource.
 - B. Shadow prices are not used to directly determine the optimal solution to the primal problem. The optimal solution is found by solving the primal problem. While related, the dual problem provides information about the sensitivity of the optimal solution.
 - C. Shadow prices are associated with constraints, not just basic variables. While basic variables play a role in determining shadow prices, the shadow prices themselves reflect the marginal value of relaxing a constraint.
 - D. Shadow prices are directly related to the dual variables. They do not represent the cost of violating a constraint; rather, they represent the value of relaxing a constraint.
 - E. Shadow prices are not calculated solely by sensitivity analysis on the primal problem. They are directly obtained from the optimal solution of the dual problem. Sensitivity analysis uses shadow prices to determine the range of changes that maintain the optimal basis.
6. In the context of shadow prices, what is the economic interpretation of a shadow price of 0 for a constraint in a linear programming problem?
 - a. A shadow price of 0 implies the constraint is not binding, not binding. A binding constraint with a non-zero shadow price would indicate that changes to its RHS would impact the objective function value.

- b. A shadow price of 0 indicates that the constraint is non-binding. This means the optimal solution already satisfies the constraint with slack, so changes to the RHS within a certain range will not affect the optimal solution or objective function value.
 - c. A shadow price of 0 indicates that the constraint is non-binding, not binding. A binding constraint would have a non-zero shadow price.
 - d. A shadow price of 0 implies the constraint is non-binding, meaning changes to its RHS within a certain range will not affect the optimal solution or objective function value.
 - e. The shadow price provides valuable information about the constraint's impact on the optimal solution. A shadow price of 0 has a specific meaning.
7. Consider a maximization linear programming problem. A non-basic variable has a shadow price of 5. What does this indicate?
- a. The shadow price represents the marginal increase in the objective function value for a one-unit increase in the resource associated with the constraint. Since it's positive, increasing the resource will improve the objective function.
 - b. Shadow prices reflect the impact of *increasing* resources, not decreasing them. A decrease might lead to a different optimal solution, but the shadow price doesn't directly quantify that.
 - c. Shadow prices are associated with constraints, not variables directly. A non-basic variable at its optimal level would have a reduced cost of 0, not a shadow price.
 - d. Shadow prices are not the values of the variables. They represent the marginal change in the objective function value per unit change in the constraint's RHS.
 - e. Shadow prices are directly relevant to the constraints and indirectly to the variables through their relationship to the constraints.
8. In a linear programming problem, the shadow price for a particular constraint is 2. What does this mean?
- a. This would be true for a minimization problem or if the shadow price were negative.
 - b. A positive shadow price in a maximization problem means that increasing the right-hand side of the constraint will increase the optimal objective function value. The shadow price gives the approximate increase per unit increase in the RHS.
 - c. A non-zero shadow price indicates that changes to the RHS will affect the objective function value.
 - d. While the basis might remain unchanged for small changes, the objective function value will change.
 - e. Shadow prices are fundamental to sensitivity analysis and directly relate to the optimal objective function value.
9. Suppose a linear programming problem has an optimal solution. If a constraint has a shadow price of 0, what does this imply?
- a. A binding constraint would have a non-zero shadow price.
 - b. A shadow price of 0 indicates that the constraint is non-binding. The optimal solution satisfies the constraint with some slack, so changes to the RHS of the constraint (within a certain range) will not affect the optimal solution or objective function value.

- c. A shadow price of 0 is associated with an optimal solution, implying feasibility.
 - d. A shadow price of 0 is associated with an optimal solution, implying boundedness.
 - e. Shadow prices are defined for all constraints in an optimal solution.
10. In a linear programming problem, a shadow price of 0 for a particular constraint implies what about the associated resource?
- a. A shadow price of 0 means the resource is not fully utilized. If it were fully utilized, its shadow price would be positive, reflecting the marginal increase in the objective function value from an additional unit of the resource.
 - b. A shadow price of 0 indicates that the resource is not a limiting factor in the optimal solution. Increasing its availability would not improve the objective function value because the optimal solution already uses less than the available amount of that resource.
 - c. A positive shadow price indicates a scarce resource. A shadow price of 0 indicates that the resource is abundant and not limiting the optimal solution.
 - d. While degeneracy can lead to shadow prices of 0, it's not the only cause. A shadow price of 0 simply means the resource is not binding in the optimal solution.
 - e. Infeasibility is unrelated to a shadow price of 0. A shadow price of 0 is a characteristic of an optimal solution where a particular resource is not binding.